Siddharth Bhat

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Summary

ML Performance Engineer at **XTX Markets**; PhD researcher (Cambridge) working at the intersection of machine learning and formal mathematics: neural proof synthesis, benchmark and evaluation design, and the symbolic machinery that makes model output checkable. Built the retrieval-augmented proof-synthesis pipeline for the F* proof assistant at Microsoft Research (**ICSE 2025 best paper**); first-authored work on **dataset defects and evaluation failures in Lean theorem-proving benchmarks** (ICML 2026); showed that a classical symbolic method beats AlphaGeometry on IMO geometry (NeurIPS 2024 MATH-AI). Awarded **two** Renaissance Philanthropy AI for Maths grants: evaluating natural-language proofs (1 of 30 groups funded from 280+ applicants) and, with EleutherAI, tactic synthesis via Lean metaprogramming. Underneath the ML: a decade of performance-critical C++/CUDA in production compilers — #3 contributor to LLVM's polyhedral loop optimizer, primary author of Lean's LLVM backend. Fluent in Python/PyTorch, Lean, C++, CUDA, Haskell, Rust.

Expertise

- **Neural theorem proving & RL (PyTorch):** designed and ran the RAG + RL pipeline that trains models to generate F* proofs, combining language-model inference with SMT-backed symbolic search (Microsoft Research; ICSE 2025 best paper); ongoing work on proof automation for Lean.
- **Evaluation & benchmarks:** first-author study of dataset defects, contamination and evaluation failures across Lean theorem-proving benchmarks (ICML 2026); established stronger symbolic baselines for AI geometry systems, showing Wu's method rivals silver medalists and lifts AlphaGeometry past gold (NeurIPS 2024 MATH-AI); co-developer of the verified solver that such benchmarks use as a proof checker.
- **Representation learning & NLP:** first-author work modelling word embeddings as tuples of feature probabilities (RepL4NLP 2020); embedding pipelines (`word2vec`, `GloVe`, `fasttext`) as a researcher and as a TA for *Natural Language: Applications*.
- **Systems for ML (C++, CUDA, LLVM/MLIR):** #3 contributor to Polly, LLVM's polyhedral loop optimizer — GPU code generation and unified-memory support for compute-intensive loop nests; stencil and tiling optimization in PolyMage/PLUTO (SC 2017); MLIR-based compiler backends targeting accelerators.

Selected Publications

1. Faults in Our Formal Benchmarking: Dataset Defects and Evaluation Failures in Lean Theorem Proving. Pawan Sasanka Ammanamanchi, *Siddharth Bhat*, Stella Biderman. *ICML 2026*.
2. Towards Neural Synthesis for SMT-Assisted Proof-Oriented Programming. Saikat Chakraborty, Gabriel Ebner, *Siddharth Bhat*, Sarah Fakhoury, Sakina Fatima, Shuvendu Lahiri, Nikhil Swamy. *ICSE 2025 (best paper)*.
3. Verifying Wu's Method can Boost Symbolic AI to Rival Silver Medalists and AlphaGeometry to Outperform Gold Medalists at IMO Geometry. Shiven Sinha, Ameya Prabhu, Ponnurangam Kumaraguru, *Siddharth Bhat*, Matthias Bethge. *NeurIPS 2024 Workshop MATH-AI*.
4. Word Embeddings as Tuples of Feature Probabilities. *Siddharth Bhat*, Alok Debnath, Souvik Banerjee, Manish Shrivastava. *RepL4NLP 2020*.
5. Certified Decision Procedures for Width-Independent Bitvector Predicates. *Siddharth Bhat*, Léo Stefanesco, Chris Hughes, Tobias Grosser. *OOPSLA 2025*.
6. Interactive Bit Vector Reasoning using Verified Bitblasting. Henrik Böving, *Siddharth Bhat*, Alex Keizer, Abdalrhman Mohamed, Léo Stefanesco, Josh Clune, Clark Barrett, Tobias Grosser. *OOPSLA 2025*.
7. Optimizing Geometric Multigrid Computation using a DSL Approach. Vinay Vasista, Kumudha KN, *Siddharth Bhat*, Uday Bondhugula. *Supercomputing (SC) 2017*.

Full list (14 papers, incl. POPL, OOPSLA, CGO, ITP) on Google Scholar.

Experience

2026–present XTX Markets, London — *ML Performance Engineer*. Performance engineering for machine-learning systems.
Jul–Sep '23 Microsoft Research, Redmond — *Research Intern*. Built a retrieval-augmented RL pipeline for automated theorem proving: PyTorch models generating F* proofs, paired with SMT-assisted search. ICSE 2025 best paper.
Sep–Nov '24 Amazon Web Services, Automated Reasoning Group, Austin — *Research Intern*. Deciding memory non-interference in a Lean-based ARM symbolic simulator; large-scale symbolic execution.
May–Jul '19 Tweag.io, Paris — *Research Intern*. Re-implemented portions of the GHC runtime for Asterius, a Haskell→WebAssembly compiler.
Mar–Dec '17 ETH Zurich, Scalable Parallel Computing — *Research Intern*. CUDA code generation in Polly for compute-intensive Fortran loop nests.
May–Jul '16 IISc Bangalore — *Research Intern*. PolyMage, a DSL compiler for data-parallel stencil computation; tiling patterns, ISL and PLUTO.

Software

Lean 4: co-developed `bv_decide`, the verified bitvector solver (5th at SMT-COMP 2025) used as a proof checker; primary author of Lean's LLVM backend.
Polly (LLVM): #3 contributor to LLVM's polyhedral loop optimizer: 121 commits, 6000 LoC C++, GPU codegen and unified memory.
lean-mlir: formal semantics for MLIR in Lean 4, machine-checking compiler rewrites.
Lean 4 Metaprogramming Book: authored the chapters on tactics and on metaprogramming for embedded DSLs.

Education & Awards

PhD, Computer Science, University of Cambridge; transferred from University of Edinburgh — 2022–2026 (writing up)
MS / BTech, Computer Science, IIIT Hyderabad, India (coursework in NLP and deep learning) — 2015–2021

Awards: Two Renaissance Philanthropy AI for Maths grants (1 of 30 from 280+ applicants; tactic synthesis, with EleutherAI) · Best paper, ICSE 2025 · 5th place, SMT-COMP 2025 (`QF_BV`) · Google Summer of Code student '15, mentor '16 and '18